# CHANDU PRASAD BHAIRAPU



**AI Engineer**

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## ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in taking AI workloads into deployment: containerized services with Docker, Infrastructure as Code with AWS CDK and Git-based CI/CD, on AWS and Microsoft Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by hands-on work on the kind of Data & AI workloads a platform has to serve, from LLM agents and RAG to classical machine learning models.

## EXPERIENCE

### Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

**Kniit GmbH**

November 2025 – present — Muldestausee, Germany

- Owned the knowledge-extraction pipeline end-to-end as sole developer (data ingestion, chunking, prompt engineering, indexing), reporting directly to the CTO and working with SAP domain experts across the German energy sector; services deployed on Azure in a Docker and Kubernetes stack.
- Developed LLM agents in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documents, and evaluated them against an SAP-system-derived ground truth test set, raising precision/recall from 33%/45% to 93%/90%.
- Built a deterministic, code-based pipeline instead of an LLM for a second, high-volume data stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed a Model Context Protocol (MCP) service that exposes the knowledge graph as a tool integration for agents, with millisecond-latency retrieval, integrated into Claude Code.
- Developed and deployed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Azure, Kubernetes, Docker, Python, FastAPI, Postgres, MongoDB, Neo4j, LangGraph, LangChain, LiteLLM, MCP, RAG, React**

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### GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

**Covestro Deutschland AG**

December 2024 – April 2025 — Leverkusen, Germany

- Advised business stakeholders as an internal consultant: mapped their decision-making workflows, identified where agentic AI could support them and translated their requirements into a solution approach.
- Built and deployed an agentic AI application with LangGraph that routes requests across vector-database retrieval, general inference and summarization.
- Containerized the stack (Python/FastAPI backend, React frontend) with Docker and deployed it to AWS with CDK as Infrastructure as Code; Git-based delivery in GitLab with CI/CD deployment pipelines.
- **Tech stack: AWS (CDK, EC2, Lambda), Docker, GitLab CI/CD, Python, FastAPI, LangGraph, LangChain, Postman, React**

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### GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

**Covestro Deutschland AG**

May 2024 – November 2024 — Leverkusen, Germany

- Defined serverless AWS infrastructure (Lambda, SQS, DynamoDB) as code with CDK for PrismaChat RAG evaluation, automating evaluation with RAGAS and PytrecEval against chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking strategies, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

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### Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

**Covestro Deutschland AG**

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models on time-series process data with engineered lag and rolling-window features, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

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### Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

**Cognizant Technology Solutions**

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

## PROJECTS

**Project: LLM Gateway | Python | FastAPI | Kubernetes (AKS) | Terraform | Microsoft Azure**

- Provider-agnostic LLM gateway as a shared, self-service service of an AI platform for Data Engineers and Data Scientists: token budgets, retries and fallback between providers, latency, token and cost logging per request; teams onboard through a configuration file.
- Containerized workload on Azure Kubernetes Service (Helm chart, health checks, autoscaling) with infrastructure in Terraform; security via Microsoft Entra Workload ID, Azure Key Vault and least-privilege RBAC, governance via Azure Policy and tags; CI/CD pipeline with tests, image scan and terraform plan/apply; runbook documentation.

**Project: Meme Generation using LLMs | Open Llama 3B | QLoRA Fine-Tuning**

- Fine-tuned an Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

## TECHNICAL SKILLSET

- **Cloud & Containers:** Microsoft Azure, AWS (EC2, Lambda, SQS, DynamoDB, S3, VPC, Load Balancer), Docker, Kubernetes
- **Infrastructure as Code, CI/CD & automation:** AWS CDK, Git, GitLab CI/CD, Git-based delivery workflows
- **Data & AI Workloads:** Python, FastAPI, Python microservices, LangGraph, LangChain, LiteLLM, RAG, PyTorch, TensorFlow, Scikit-Learn, RAGAS
- **Data:** SQL, Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search), Pandas
- **Ways of working:** collaboration with domain experts and stakeholders, AI-assisted development with Claude Code, Agile

## EDUCATION

**Master of Science in Artificial Intelligence** — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

**Bachelor of Technology in Mechanical Engineering** — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

## PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

## LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)